



Classifying Sleep Stages Using Bitbrain Open Access Sleep (BOAS) Data

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Background

- EEGs non-invasively measure electrical brain activity by placing electrodes on the scalp to detect voltage potentials generated by neuronal current flow.
- EEGs are useful in linking voltage signals to neurological network changes.
- ML models trained on EEG data can decode distinct brain states, enabling applications in brain-computer interfaces and neurological therapies.
- Sleep consists of 4 stages (Stage 1, Stage 2, Stage 3 slow-wave, and REM), each defined by distinct patterns of brain activity, muscle tone, and eye movement, with a full cycle lasting approximately 90 minutes.
- Manual sleep staging is time-intensive; automated ML classification could enable scalable, wearable-based diagnosis of sleep disorders.

Research Goal

To develop a machine learning model that classifies EEG data into five classes (Wake, N1, N2, N3, REM) by using distinctive EEG features such as theta activity, sleep spindles, K-complexes, delta waves, and REM-associated patterns.

Methods

- Dataset: OpenNeuro ds005555; 128 subjects with overnight polysomnography (PSG) and wearable headband EEG recordings
- Extracted HB_1 headband EEG channel
- Segmented continuous EEG into 30-second epochs aligned to PSG annotations
- Subject-level split: 85% train, 15% validation
- Inverse-frequency class weights in CrossEntropyLoss to address class imbalance
- 15 epochs, batch size = 64; best checkpoint saved by peak validation accuracy

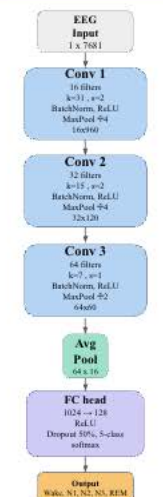


Figure 1: 1-D CNN Architecture

Class Model Training vs Validation Accuracy



Figure 2: 5-Class Model Training and Validation Accuracy. Peak performance was achieved at epoch 13, with a training accuracy of 71.35% and validation accuracy of 66.82%. The divergence between training and validation accuracy in later epochs suggests the onset of overfitting.

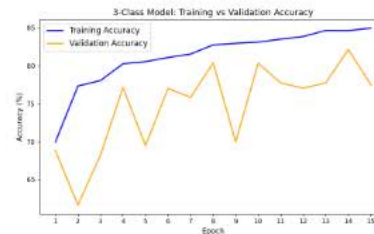


Figure 3: 3-Class Model Training vs Validation Accuracy. The model achieved improved performance compared to the 5-class model, with a peak validation accuracy of 82.16% and training accuracy of 84.64% at epoch 14/15, indicating that merging NREM substages enhances model performance.

Confusion Matrix

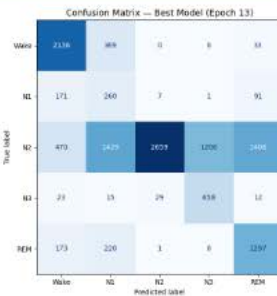


Figure 4: 5-Class Model Confusion Matrix at the best epoch. The 5-class model shows a weaker diagonal pattern, with substantial misclassification concentrated in N1 and N2, where frequent confusion with adjacent stages reflects the inherent similarity of transitional stage EEG signatures.

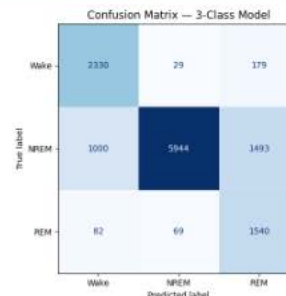


Figure 5: Simplified 3-Class Model Confusion Matrix. The 3-class model shows a strong diagonal pattern across all three stages, reflecting substantially improved classification over the 5-class model, with residual confusion concentrated at the Wake-NREM and NREM-REM boundaries.

Accuracy at Best Epoch

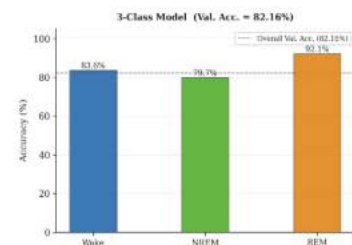


Figure 6: 3-Class Model Per-Class Accuracy. Per-class accuracy: Wake 83.6%, NREM 79.7%, and REM 92.1%. NREM precision of 0.984 confirms that consolidating transitional stages reduced inter-class confusion.

Cross Analysis

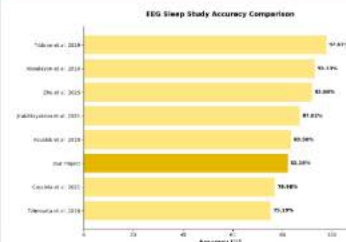


Figure 7: Cross-Analysis of Model Accuracy. Comparison of accuracy across published EEG-based sleep staging studies show that our model (82.16%) achieves competitive performance despite using a simplified, single-channel approach.

Discussion

- Performance varied by stage, with N3 and REM classified most accurately due to distinct EEG patterns, while N1 and N2 remained slightly above chance due to subtle transitions and overlapping features.
- Simplifying the task to 3 classes (Wake/NREM/REM) significantly improved performance, suggesting that reduced granularity may be more practical for real-world applications with current technologies.
- Comparison with published EEG sleep staging studies (Figure 7) suggests that multi-channel approaches tend to yield higher classification accuracy, highlighting a tradeoff between model performance and the simplicity of single-channel wearable systems.
- The use of headband EEG (HB_1 only) shows that reliable sleep classification is achievable without full polysomnography, supporting the potential for wearable, at-home sleep monitoring outside clinical settings.

Future Directions

- To improve accuracy in classification of lighter sleep stages, specifically N1 and N2, we can incorporate multimodal data and additional datasets.
- Future work will evaluate model generalizability across diverse populations, recording conditions, and wearable EEG platforms via cross-validation and external datasets testing, which will be critical for detecting and monitoring sleep disorders in applied clinical settings.

References

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